International Journal of Hospitality Management 134 (2026) 104602

Contents lists available at ScienceDirect

International Journal of Hospitality Management

journal homepage: www.elsevier.com/locate/ijhm

Full length article

Unintended prosocial consequences of robot prevalence: The mediating role of realistic threat

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ARTICLE INFO

Keywords:
Service robots
Robot workforce
Realistic threat
Prosocial behavior
Experimental studies

ABSTRACT

With the growing shift to service automation in the hospitality industry, robot prevalence is on the rise. Drawing on the intergroup threat theory and terror management theory, we argue that as service robots become more prevalent, consumers may become prosocial in response to a realistic threat. Additionally, we theorize that these effects will be salient among those who are employed, especially those who use artificial intelligence or robots at work. The results of four experimental studies support our hypotheses. Our findings provide practical insight into how companies can leverage such behavioral shifts among consumers to maximize the effectiveness of their prosocial initiatives.

1. Introduction

Recently, the hospitality service robot market has been experiencing significant growth because of increasing demand for automation, enhanced customer experience, and operational efficiency in the hospitality industry. Labor shortages in the hospitality industry have accelerated the adoption of service robots, with global sales reaching over 205,000 units in 2023—a 30 % increase from the previous year—and continued double-digit growth projected through 2026, particularly in sectors such as hospitality, logistics, and cleaning (Müller et al., 2024). With the growth of the robot workforce across industries, the news media has given increased attention to how it could affect humanity, often framing it negatively (e.g., Time, 2023).

The increasing integration of service robots has generated a large body of literature on consumers' and employees' adoption and interaction with service robots in the hospitality field (e.g., Begum et al., 2025; Leung et al., 2023; Song, Zhang, and Wu, 2022). In particular, a line of research has explored how the growing presence of service robots elicits a sense of threat among human service employees, thereby influencing service performance (Xu et al., 2025; Yu et al., 2026; Zhang et al., 2025). However, service employees are note the only ones who feel threaten by this shift; the prevalence of service robots in workforce poses threats to broader stakeholders as well, including consumers, potentially influence their behaviors (Jackson, Castelo, and Gray, 2020). Nevertheless, few studies explore how the rise of the robot workforce is resulting in unintentional consequences for consumers. To fill this gap, this research examines the unintended ripple effects of a rising robotic workforces on consumers' prosocial behavior. We focus on consumers' prosocial behavior, which refers to any action of benefiting others, not only because of its critical social implications but also because it is vital for maintaining the communal atmosphere essential to hospitality settings (Ali et al., 2024; Boo, Kim, and Kim, 2024; Jia et al., 2024; Kang and Baker, 2022). While a growing body of work examines the link between AI, robots, and prosocial behavior, it has largely focused on two primary scenarios: robots directly eliciting prosocial actions from consumers (Huang et al., 2024; Lo et al., 2022; Sadiq et al., 2024; Xu et al., 2025), or consumers performing prosocial acts directed *toward* the robots themselves (Granulo et al., 2024; Liu et al., 2023; see Oliveira et al., 2021 for a review). Extending this line literature, our research identifies robot prevalence as a seemingly unrelated situational cue, which can exert an unintended impact on consumer prosocial behavior.

Building on intergroup threat theory (Stephan, Ybarra, and Rios, 2015), we argue that consumers experience a realistic threat—the threat of losing their jobs and, thus, material resources—from robots. Based on terror management theory (Greenberg, Pyszczynski, and Solomon, 1986), we further argue that individuals could become more prosocial as a means to cope with the threat induced by the prevalence of robots. By focusing on prosociality, we aim to reveal how the robotic threat unintentionally catalyzes a search for human connection and social value (Ali et al., 2024; Jia et al., 2024). Across four experimental studies, we

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https://doi.org/10.1016/j.ijhm.2026.104602
Received 23 February 2025; Received in revised form 24 January 2026; Accepted 9 February 2026
Available online 12 February 2026
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demonstrate that consumers tend to donate more, help other people more, and behave more pro-environmentally when exposed to the prevalence of robots. We also find that this effect on prosocial behavior is mediated by realistic threat. To validate that realistic threat is indeed the driver of this behavior, we examine a boundary condition that expected to mitigate or amplify the perception of threat: the use of artificial intelligence (AI) and robots at work. The findings suggest that the proposed effects are particularly salient among employed individuals, especially those currently using AI and robots. These moderation effects further demonstrate the robustness of realistic threat mechanism, showing that groups who are more relevant to robot prevalence are more prone to prosocial behavior.

This research contributes to the service robotics literature by identifying a novel and unintended downstream consequence of robot prevalence in the hospitality industry: an increase in consumer prosociality. That is, merely being in an environment where service robots are widely adopted to replace the human workforce prompts consumers to engage more readily in prosocial behavior. Moreover, we find that realistic threats can partly explain the effect of robot prevalence on prosocial behavior. Finally, we examine the robustness of the role of realistic threats and demonstrate that individuals affected by AI and robots exhibit increased prosocial behavior. Therefore, our findings deepen understanding of consumer behavior in the era of service robots and social well-being.

From a managerial perspective, businesses in the hospitality industry may gain an advantage by learning about the unintended effects of robot prevalence. This will allow them to create more tailored service offerings for their customers. Our findings suggest that companies should offer various prosocial initiatives in the robot era to increase consumer participation and have a positive social impact. Furthermore, our findings show that prosocial initiatives targeting employed people, especially those affected by AI/robots at work, should be more effective.

2. Theoretical background

2.1. The prevalence of the robot workforce can pose a realistic threat

The literature documents that humans feel threatened by the rise of robots (e.g., Huang et al., 2021; Złotowski et al., 2017). Robots can elicit an innate fear of being replaced, loss of control, and even death, provoking in individuals a sense of uncanniness and a perceived threat to human identity (MacDorman and Ishiguro, 2006; Ray, Mondada, & Siegwart, 2008; Xu et al., 2024). Given this perceived threat, consumers tend to prefer human service employees or non-humanoid service robots to humanoid service robots in service encounters (Kim, Lee, and Kang, 2023; Liu, Wan, and Yi, 2022; Singh, Olson, and Tsai, 2021). Moreover, when interacting with humanoid robots, individuals are likely to engage in compensatory behaviors (e.g., purchasing status goods and eating more food) in response to a threatening experience (Mende et al., 2019).

In this research, drawing on intergroup threat theory (Stephan et al., 2015), we theorize that the prevalence of robots can pose a realistic threat. This theory posits that it is human nature to categorize social beings into ingroups and outgroups and, often, to regard an outgroup as a threat (Branscombe et al., 1999; Tajfel and Turner, 1986). Intergroup threat theory includes two types of threats: realistic and symbolic threats (Stephan et al., 2015). A realistic threat is a concern about a loss of material resources, safety, or physical well-being of the ingroup, whereas a symbolic threat is a concern about the integrity or validity of the ingroup's meaning system (e.g., religion, philosophy, or morality) (Stephan et al., 2015). We focus on realistic threats in this research as it is a more tangible, direct and universal risk to society and human beings, while the symbolic threat is a more intangible, indirect, and personal risk to individuals and group distinctiveness. Therefore, realistic threat has been found to trigger stronger and immediate responses to customers than symbolic threat (Xu et al., 2024).

With the rise of the robot workforce, individuals are likely to categorize all of humankind as an ingroup and robots as an outgroup (Jackson et al., 2020). As robots become increasingly intelligent and capable, they pose a realistic threat of taking humans' jobs and, thus, their economic resources (Granulo, Fuchs, and Puntoni, 2019; Li et al., 2024). Some robots can outperform human employees on physical and cognitive tasks (Frey and Osborne, 2017; Huang and Rust, 2021). Furthermore, people tend to feel a stronger realistic threat when robots possess more humanlike features (Huang et al., 2021; Wang, Kim, and Kim, 2023). Although the new generation of robots is not necessarily replacing human employees but is augmenting their abilities and efficiencies, individuals may nevertheless fear that their employment and welfare are at risk (Wu and Zhang, 2024; Yam et al., 2023; Yogeeswaran et al., 2016). Such a perceived threat results in various negative consequences, such as reduced self-esteem (Spatola and Normand, 2021), task underperformance (Tang et al., 2023a), job burnout and workplace incivility behaviors (Yam et al., 2023), workplace withdrawal behaviors (Pan, Lin, and Wong, 2025), and increased insomnia and alcohol consumption (Tang et al., 2023b).

2.2. How realistic threats trigger prosocial behaviors

Previous research on threat and defense posits that individuals who experience threats tend to activate coping strategies to reduce negative feelings (e.g., anxiety) and retain control, meaning, or belonging (Jonas, Sullivan, and Greenberg, 2013; Uenal et al., 2021). A body of research establishes that prosocial behavior can buffer stress while increasing happiness and life satisfaction (Aknin et al., 2013; Keyes and Haidt, 2003). For example, in the context of the COVID-19 pandemic, Hu, He, and Zhou (2020) demonstrated that pandemic-related threats increased employee anxiety, which in turn motivated greater engagement in prosocial behaviors (e.g., volunteering). Furthermore, their research revealed that servant leadership can amplify this tendency. By fostering compassion and elevating employees' feelings of self-worth, servant leadership provides the psychological resources that enable individuals to better manage existential threats, a mechanism consistent with the principles of terror management theory (Greenberg and Arndt, 2012; Hu et al., 2020).

Besides, terror management theory (Greenberg et al., 1986) suggests that anxiety stemming from threats leads individuals to maintain their cultural worldviews and protect their self-esteem (Hirschberger, Ein-Dor, and Almakias, 2008; Jonas et al., 2002). Consequently, engaging in prosocial behavior can serve as an effective terror management mechanism, particularly when such actions are culturally valued and bolster an individual's sense of self-worth (Ferraro, Shiv, and Bettman, 2005; Jonas et al., 2002). Empirical research corroborates this, demonstrating that when individuals experience threats, they are more likely to help others (e.g., through donations), especially when the beneficiary is an in-group member (Jonas et al., 2002), when virtue is a central component of self-esteem (Ferraro et al., 2005), and when prosocial norms are salient (Cai and Wyer, 2015; Jonas et al., 2008).

Moreover, facing a collective threat (e.g., terrorism, unemployment, a pandemic), people tend to reaffirm their social group and to share a group identity with a relevant social category to restore a sense of control (Fritsche, Jonas, and Kessler, 2011). Most relevant to our work, Jackson, Castelo, and Gray (2020) find that the rise of a robot workforce can reduce prejudice toward outgroups by enhancing the perception of a common human ingroup. That is, the rise of robots prompts people to redefine their sense of community by emphasizing the commonalities among humans, regardless of race or nationality (i.e., 'We are all humans'), while highlighting the differences between humans and non-human groups (i.e., robots) (Gaertner et al., 1993; Jackson et al., 2020). Therefore, people may identify themselves with humanity as a whole and be concerned about a global humanitarian issue, such as job loss (McFarland, Webb, and Brown, 2012).

In summary, we argue that the increasing presence of robots in the workforce is likely to encourage consumers to engage in more prosocial

behaviors. This heightened prosociality arises from the perceived realistic threat posed by a growing number of robots replacing human jobs. Thus, we formally hypothesize the following:

H1: The prevalence of the robot workforce promotes consumers' prosocial behaviors.

H2: The relationship between the prevalence of the robot workforce and consumers' prosocial behavior is mediated by realistic threat.

2.3. Boundary condition: employment status and ai/robot use at work

As our main arguments pertain to the realistic threat posed by robots (i.e., the threat of taking humans' economic resources), we propose that certain groups of individuals may respond more strongly to robot prevalence, hence increasing their propensity for prosocial behaviors. That is, individuals who are employed may be more influenced by the rise of robots than those who are unemployed. Employed individuals may be under stress and fear being replaced by robots depending on their level of advancement in various fields (Pan et al., 2025; Wang et al., 2023; Zhao, Yuan, and Song, 2024). This is particularly relevant for individuals who are using any forms of AI, including robots, in their jobs (Chen and Cai, 2025). Therefore, employed individuals, particularly those who use AI or robots at work, could perceive a greater realistic threat from robot prevalence and thus become more prosocial. We therefore hypothesize the following.

H3a. Employment status moderates the positive relationship between the prevalence of the robot workforce and consumers' prosocial behavior, such that this positive relationship is stronger for those who are employed.

H3b. Among those who are employed, the positive relationship between the prevalence of the robot workforce and consumers' prosocial behavior is particularly stronger for those who use AI/robots more at work.

3. Overview of studies

This study employed online experiments to test our hypotheses. In Study 1, we first established causality regarding the hypothesized main effect of robot prevalence on consumer prosocial behavior. Then, Studies 2a and 2b examined the mediating role of realistic threat in the robot prevalence effect on prosocial behavior. In these studies, we demonstrated the robustness of the mechanism (i.e., realistic threat) by ruling out alternative explanations (i.e., symbolic threat and emotions in Study 2a and 2b, respectively). Finally, Study 3 showed that the effect of robot prevalence is particularly evident among individuals who are employed, especially those who work with AI or robots (Fig. 1).

We applied various operationalization of consumer prosocial behavior (adapted from Penner et al., 2005) throughout the studies to maximize generalizability of the findings: helping strangers (Study 1), helping other customers (Study 2a), donation (Study 2b), and pro-environmental behavior (Study 3).

3.1. Study 1. experimental evidence of the main effect

3.1.1. Method

One hundred participants (M_{age} = 38.16, 43.0 % female) were recruited from Prolific online consumer panel. They were randomly assigned to a robot-prevalence condition or a control condition. In the robot condition, the participants were asked to watch a 1-minute 3D animation depicting a fictitious restaurant where robots served customers. In the control condition, a similar 3D animation depicting the same restaurant but staffed with human employees was presented. The participants were asked to picture themselves visiting the restaurant (see Appendix for details). Then, the participants were given six hypothetical scenarios in which they had the opportunity to help strangers (e.g., giving money to a homeless person; adapted from DeWall et al., 2008) and indicated their likelihood of helping in each situation on a 9-point scale (1 = "not at all likely," 9 = "very likely"; α =.72).

A separate pretest was conducted to validate the 3D animation video stimuli. Seventy-seven participants (M_{age} = 38.12, 48.1 % female), recruited from the same pool in Study 1, were randomly assigned to a robot-prevalence condition or a control condition. They were presented with the same 3D animations depending on their assigned condition. After viewing the assigned video, participants completed a manipulation check by rating their agreement with the statement that the hotel's service was automated (1 = strongly disagree, 7 = strongly agree). They also assessed anticipated service quality of the restaurant ("I expect the service quality of the restaurant to be …", 1 = low/poor/unable to meet my expectation, 7 = high/excellent/able to meet expectation; α =.94). Independent *t*-tests showed that participants in the robot-prevalence condition are more agreed (M = 6.67, *SD* =.62) that the restaurant service is automated by service robots than those in the control condition (M = 3.29, SD = 2.35; $t(75) = 8.68$, $p < .01$). However, they anticipated a similar level of service quality ($M_{robot\text{-}prevalence} = 5.51$, $SD_{robot\text{-}prevalence} = 1.35$ vs. $M_{control} = 5.58$, $SD_{control} = 1.28$; $t(75) < .01$). These results confirm that the video stimuli successfully manipulated perceptions of robot prevalence without creating a confounding effect on service quality expectations.

3.1.2. Results

Likelihood of helping strangers. An independent samples *t*-test revealed a significant main effect of robot prevalence manipulation on the participants' likelihood of helping strangers ($t(98) = 2.22$, $p < .05$, Cohen's $d = .45$). As hypothesized, the participants in the robot condition indicated a greater likelihood of helping (M = 6.68, SD = 1.23) than those in the control condition (M = 6.07, *SD* = 1.48; Fig. 2).

3.1.3. Discussion

Study 1 provides preliminary causal evidence that consumers who visualized a restaurant where service robots replaced human agents

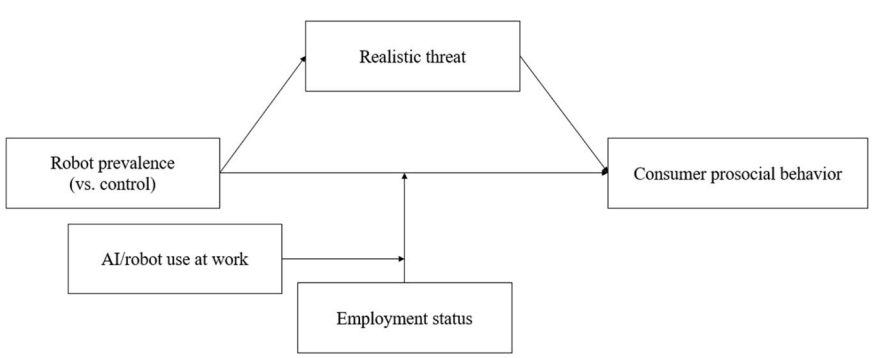

Fig. 1. Conceptual Framework.

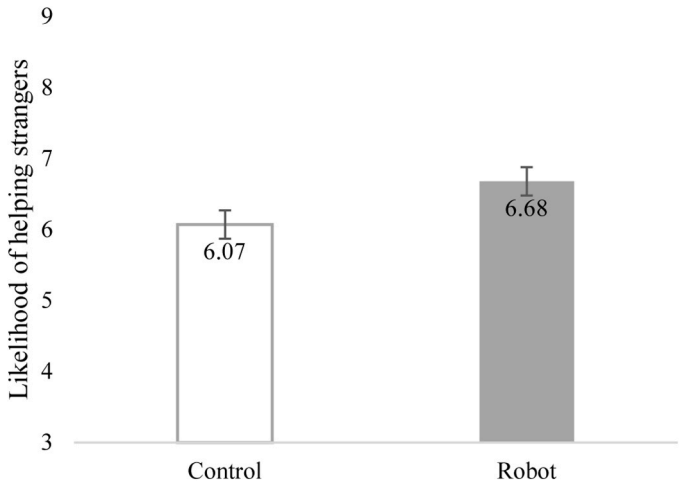

Fig. 2. Likelihood of Helping Strangers in Study 1.

were more likely to engage in prosocial behaviors—specifically helping strangers—than those who visualized a restaurant staffed by humans.

However, a post-hoc power analysis using G*Power (Faul et al., 2009) revealed that given the final sample size and the effect size, we achieved a power of.70, which is below the conventional power threshold of.80 in behavioral studies (Cohen, 2013). Thus, in the subsequent studies, we conducted *a priori* power analyses (targeting a medium effect size (Cohen's d =.50) with 80 % power) and obtained sufficient sample sizes.

3.2. Study 2a. test of the psychological mechanism: realistic threat, ruling out symbolic threat as an alternative mechanism

3.2.1. Method

An *a priori* power analysis indicated that a sample of at least 128 was necessary to detect a medium effect size with 80 % power. Thus, we recruited 240 participants (M_{age} = 38.95, 47.9 % female) from Prolific online consumer panel. The participants were randomly assigned to either a robot-prevalence condition or a control condition. The participants were asked to visualize themselves visiting a restaurant where many parts of frontline service tasks are automated by service robots (the robot-prevalence condition) or they are carried out by human agents (the control condition). Then, they were asked to describe what their experience would be and how they would feel during their stay at the hotel (see Appendix for details). After the writing task, the participants were asked to indicate their willingness to help other customers at the restaurant (e.g., "I will assist other customers of the restaurant if they need help." 1 = strongly disagree, 7 = strongly agree; α =.89; Assiouras et al., 2019) and help the human employees (e.g., "If the employee of the restaurant makes a mistake during service delivery, I will be willing to show patience." 1 = strongly disagree, 7 = strongly agree; α =.75; Assiouras et al., 2019). Next, the participants rated realistic threat on three items (e.g., "The increased use of robots in our everyday life is causing more job loss for humans" α =.79; Złotowski et al., 2017) and symbolic threat on four items (e.g., "Recent advances in robot technology are challenging the very essence of what it means to be human" α =.88; Złotowski et al., 2017) scored on a 7-point scale. Finally, we conducted a manipulation check by asking participants whether the restaurant the visualized provides automated service delivery by robots (1 = strongly disagree, 7 = strongly agree).

3.2.2. Results

Manipulation check. An Independent t-test showed that participants in the robot-prevalence condition are more agreed (M = 6.08, SD =.95) that the restaurant service is automated by service robots than those in the control condition (M = 2.38, SD = 1.42; $t(238) = 23.69$, $p < .01$). Therefore, the manipulation of Study 2a was successful.

Willingness to help other customers and employees. An independent samples t-test showed that the robot prevalence manipulation had a significant main effect ($t(238) = 3.06$, $p < .01$, Cohen's d = .40). As hypothesized, the participants in the robot-prevalence condition were more willing to help other customers of the restaurant (M = 4.38, SD = 1.31) than those in the control condition (M = 3.85, SD = 1.39).

Another t-test showed that the robot prevalence manipulation had a significant main effect ($t(238) = 2.36$, $p < .05$, Cohen's d = .30). Unexpectedly, the participants in the robot-prevalence condition were less willing to help the employees of the restaurant (M = 4.48, SD = 1.12) than those in the control condition (M = 4.82, SD = 1.14).

Realistic and Symbolic Threat. Independent samples t-tests revealed that the robot prevalence manipulation had a significant main effect on realistic threat ($t(238) = 4.31$, $p < .01$, Cohen's d = .56) and symbolic threat ($t(238) = 3.71$, $p < .01$, Cohen's d = .48). The participants in the robot condition perceived a greater realistic threat and symbolic threat than those in the control condition (realistic threat: $M_{robot\text{-}prevalence}$ = 4.51, $SD_{robot\text{-}prevalence}$ =.99 vs. $M_{control}$ = 3.92, $SD_{control}$ = 1.14; symbolic threat: $M_{robot\text{-}prevalence}$ = 4.47, $SD_{robot\text{-}prevalence}$ = 1.40 vs. $M_{control}$ = 3.75, $SD_{control}$ = 1.59).

Mediation analysis. We ran a mediation analysis with 10,000 bootstrap samples (Model 4, Hayes, 2022), entering the robot prevalence versus control as the independent variable, willingness to help other customers as the dependent variable, and perceived realistic and symbolic threats as the parallel mediating variables. The results showed that perceived realistic threat mediated the effect of robot prevalence manipulation on willingness to help other customers (indirect effect =.19, SE =.10, 95 % CI =.03,.40), while symbolic threat did not mediate the effect (indirect effect = −.04, SE =.07, 95 % CI = −.19,.10).

3.2.3. Discussion

Study 2a provides additional causal evidence that consumers who visualized a restaurant where service robots replaced human agents were more likely to engage in prosocial behaviors (i.e., helping other customers) than those in the control condition. We found that robot prevalence manipulation can trigger both realistic and symbolic threats by service robots; however, it is perceived realistic threat that explains the enhanced consumers' prosocial behavior given robot prevalence, not symbolic threat. These results strengthen our arguments that realistic threat plays a more pivotal role in explaining the unintended prosocial consequence of robot prevalence than symbolic threat.

3.3. Study 2b. test of the psychological mechanism: realistic threat, ruling out emotions as an alternative mechanism

3.3.1. Method

An a priori power analysis indicated that a sample of at least 128 was necessary to detect a medium effect size with 80 % power. Thus, we recruited 203 participants (M_{age} = 35.31, 48.3 % female) from Prolific online consumer panel. The participants were randomly assigned to either a robot condition or a control condition. The participants were asked to visualize themselves visiting a hotel where service robots or human agents provide frontline hotel services. Then, they were asked to describe what their experience would be and how they would feel during their stay at the hotel (see Appendix for details). After the writing task, the participants rated realistic threat on the same measurement scale used in Study 2a (α =.84; Złotowski et al., 2017). Next, the participants completed an ostensible filler task, indicating their current emotions on the PANAS measurement scale (1 = "not at all," 5 = "extremely"; Watson, Clark, and Tellegen, 1988), which consists of 10 positive (e.g., "interested"; α =.92) and 10 negative (e.g., "upset"; α =.91) emotions. Emotions were tested as a potential alternative mechanism as previous research has shown a significant association between presence of service robots and emotional responses (e.g., Gong, 2025) and between emotions and prosocial behavior (e.g., Argadian and Paramita, 2025;

Lefebvre, Boman, and Orlowski, 2024). After the filler task, the participants were notified that one individual would be randomly chosen to receive a bonus payment of US$100. If selected, they would have the option to receive the entire $100 in cash or to donate a portion to a charity that assists individuals living in poverty globally. The participants specified their donation amount on a slider from $0 to $100.

3.3.2. Results

Willingness to donate. An independent samples *t*-test showed that the robot prevalence manipulation had a significant main effect ($t(201) = 2.63$, $p < .01$, Cohen's $d = .37$). As hypothesized, the participants in the robot condition were willing to donate more (M = 37.20, SD = 31.40) than those in the control condition (M = 26.40, *SD* = 26.82).

Realistic threat. An independent samples *t*-test showed that the robot prevalence manipulation had a significant main effect on realistic threat ($t(201) = 4.67$, $p < .01$, Cohen's $d = .66$). As hypothesized, the participants in the robot condition perceived a greater realistic threat (M = 4.89, SD = 1.50) than those in the control condition (M = 3.94, *SD* = 1.36).

PANAS. Independent samples *t*-tests revealed that neither positive nor negative emotions were affected by the agent type manipulation ($t_{positive}(201) = 1.13$, $p > .25$, Cohen's $d = .16$ and $t_{negative}(201) = .05$, $p > .95$, Cohen's $d = .01$).

Mediation analysis. We ran a mediation analysis with 10,000 bootstrap samples (Model 4, Hayes, 2022). As predicted, perceived realistic threat mediated the effect of robot prevalence manipulation on willingness to donate (indirect effect = 4.65, *SE* = 1.76, 95 % CI = 1.61, 8.51). As a robustness check, we added positive and negative emotions as parallel mediators (along with realistic threat). The results showed that neither positive (indirect effect =.87, *SE* =.84, 95 % CI = −.68, 2.68) nor negative (indirect effect =.004, *SE* =.28, 95 % CI = −.58,.63) emotions mediated the effect, but realistic threat still did (indirect effect = 3.87, *SE* = 1.63, 95 % CI = 1.02, 7.47; Fig. 3).

3.3.3. Discussion

Study 2b provides additional causal evidence that consumers who visualized a hotel where service robots replaced human agents were more likely to engage in prosocial behaviors—specifically donating to a charity—than those who visualized a hotel staffed by humans. In addition, the findings showed that a perceived realistic threat can at least partially explain the enhanced prosocial behavior given robot prevalence while ruling out a potential alternative explanation (i.e., positive and negative emotions).

3.4. Study 3. test of boundary conditions: employment status and AI/robot use at work

3.4.1. Method

We used a 2 (robot prevalence vs. control; manipulated) by 2 (employment status: employed vs. unemployed; measured) quasi-experimental design. We used a stratified sampling method to balance the sample sizes of different employment statuses. One hundred and ninety-nine participants (M_{age} = 35.31, 48.3 % female) recruited from Prolific were randomly assigned to the human condition or the robot condition. An *a priori* power analysis suggests that a sample of at least 128 is needed to detect a medium effect size (Cohen's f =.25) with 80 % power in a four-group design. Thus, our sample size was sufficient.

We first measured the participants' demographic variables, including employment status (1 = employed full-time, 2 = employed part-time, 3 = unemployed and looking for work, 4 = unemployed and not looking for work, 5 = retired, 6 = student). Those who were employed full-time or part-time were instructed to indicate their daily AI/robot usage at work with three items scored on a 7-point scale (e.g., "I use AI/robots to carry out most of my job functions."; α = 90.; Tang et al., 2023a). Then, the participants were asked to watch a 1-minute 3D animation depicting a fictitious hotel where either robots or human employees were serving customers and to picture themselves visiting the hotel. Next, the participants were asked how likely they would be to engage in pro-environmental behaviors during a stay at the imagined hotel on three items scored on a 7-point scale (e.g., "How likely are you to save water and energy while staying at the hotel?"; α =.94; Tussyadiah and Miller, 2019).

Like Study 1, a separate pretest was conducted to validate the 3D animation video stimuli. Seventy-five participants (M_{age} = 37.33, 50.7 % female) were randomly assigned to a robot-prevalence condition or a control condition. After viewing the assigned video clip, participants completed a manipulation check by rating their agreement with the statement that the hotel's service was automated (1 = strongly disagree, 7 = strongly agree) and anticipated service quality of the hotel, using the same measurements in Study 1 (α =.95). Independent *t*-tests showed that participants in the robot-prevalence condition are more agreed (M = 6.58, *SD* =.68) that the hotel service is automated by service robots than those in the control condition (M = 3.41, *SD* = 2.02; $t(73) = 9.16$, $p < .01$). However, they anticipated a similar level of service quality ($M_{robot\text{-}prevalence} = 5.22$, $SD_{robot\text{-}prevalence} = 1.51$ vs. $M_{control} = 5.66$, $SD_{control} = 1.29$; $t(73) = 1.35$, $p = .18$). These results confirm that the video stimuli successfully manipulated perceptions of robot prevalence.

3.4.2. Results

Pro-environmental behavior intention. A two-way ANOVA indicated a significant interaction effect between the robot prevalence manipulation and employment status ($F(1, 195) = 7.81$, $p < .01$, Cohen's f = .19; Fig. 4). Pairwise comparisons showed that the participants who were employed indicated higher pro-environmental behavior intention in the robot condition (M = 3.87, *SD* = 1.92) than in the control condition (M = 3.15, *SD* = 1.68; $F(1, 195) = 4.69$, $p < .05$). However, the participants who were unemployed indicated marginally higher pro-

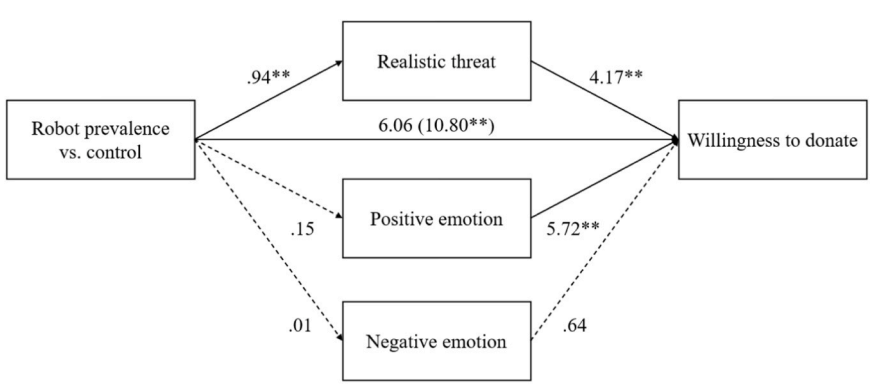

Fig. 3. Mediation Analysis in Study 2b. Note. * $p < .05$, ** $p < .01$; on the middle path, the value in parentheses is the total effect, and the value outside the parentheses is the direct effect.

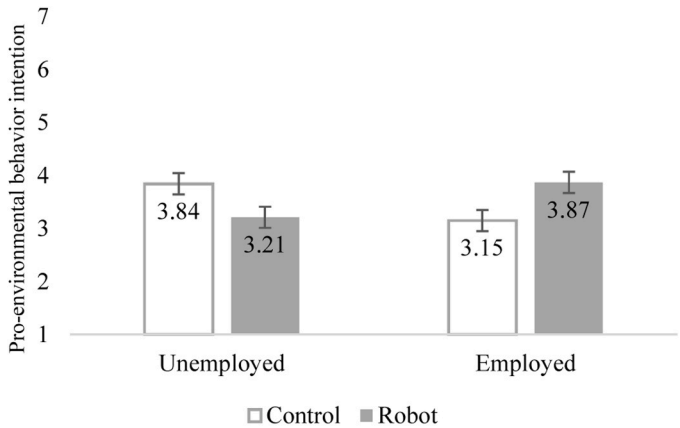

Fig. 4. Interaction Effect on Pro-environmental Behavior Intention in Study 3.

environmental behavior intention in the control condition (M = 3.84, *SD* = 1.47) than in the robot condition (M = 3.21, *SD* = 1.69; $F(1, 195) = 3.24$, $p = .07$).

Impact of AI/robot usage at work. To obtain a more detailed picture, we further examined the impact of AI/robot usage at work on pro-environmental behavior intention only for those who were employed (n = 105). The results of a moderation analysis (PROCESS Model 1; Hayes, 2020) showed a marginally significant interaction between the robot prevalence manipulation and AI/robot usage at work ($b = .33$, $SE = .20$, $t(101) = 1.69$, $p = .09$). We used a floodlight analysis to decompose the interaction and explore the Johnson-Neyman region of significance (see Fig. 5; Spiller et al., 2013). As predicted, the effect of the robot prevalence manipulation was significant among those who scored 3.72 or above on the AI/robot usage scale, namely those who use more AI/robot at work ($b = .76$, $SE = .38$, $t(101) = 1.98$, $p = .05$, 95 % CI = .00, 1.52), while the effects were not significant among those who scored lower than 3.72 (i.e., those using AI/robot at work less).

3.4.3. Discussion

Study 3 showed that the participants who visualized a hotel where service robots replaced human agents were more likely to engage in prosocial behaviors—specifically pro-environmental behaviors—than those who visualized a hotel staffed by humans. Moreover, we found that such effects of robot prevalence were prominent among those who were employed, particularly those who used AI/robot at work more.

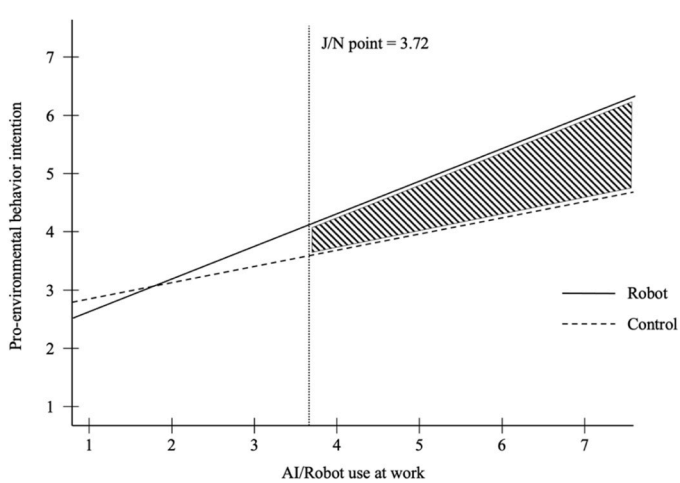

Fig. 5. Floodlight Analysis: Interaction Effect of AI/Robot Use at Work and Robot Prevalence Manipulation on Pro-environmental Behavior Intention in Study 3.

4. General discussion

Through four studies, we found that consumers were more likely to engage in prosocial behavior given greater robot prevalence. Moreover, the results showed that perceived realistic threat mediated the robot prevalence effect on consumer prosocial behavior (Studies 2a and 2b). Furthermore, the positive effect of robot prevalence on consumer prosocial behavior was salient among those who were employed, especially those who used AI/robots at work more (Study 3).

We applied (1) different approaches for robot prevalence manipulation (i.e., Studies 1 and 4: 3D animations; Studies 2 and 3: visualization tasks), (2) different contexts (i.e., Studies 1 and 2: restaurant; Studies 3 and 4: hotel), and (3) different manifestations of prosocial behavior across the studies (i.e., Studies 1 and 2a: helping others; Study 2b: making donations; Study 3: pro-environmental behavior) and observed consistent results, which highlight the generalizability of the proposed effects.

4.1. Theoretical contribution

With this research, we contribute to a growing area of study on the unintended consequences of the rise of robots. While previous studies on service robots largely focus on consumers' adoption of service robots (e.g., Begum et al., 2025; Leung et al., 2023; Song, Zhang, and Wu, 2022), scholars have recently begun to examine the unintended downstream effects of robot prevalence in the service industry, including negative workplace behaviors by service employees (e.g., Pan et al., 2025; Yam et al., 2023) and compensatory consumption by customers (e.g., Choi, Wan, and Mattila, 2024; Mende et al., 2019). Given the increasing demand for social well-being, recent studies explore the unintended social impacts of robot prevalence, such as intergroup relations (Jackson et al., 2020) and pro-environmental behaviors (Guan et al., 2024). Our research expands this line of research by providing experimental evidence of how robot prevalence influences individuals' prosocial behaviors. Particularly, our findings are consistent with the previous literature showing that the unintended positive effects of the prevalence of robot workforce on prosocial behaviors (Guan et al., 2024; Jackson et al., 2020; Xu et al., 2025).

In a similar vein, our work adds to the hospitality literature on consumer prosocial behavior. Previous research in this area mainly demonstrates the effects of managerial interventions (e.g., appeals, message framing) on boosting tourists' prosocial behavior (e.g., Boo, Kim, and Kim, 2024; Lei, Huang, and Yang, 2024). Another body of literature shows that service robots could play an important role in persuading consumers to engage in prosocial behaviors (Huang et al., 2024; Lo et al., 2022; Sadiq et al., 2024; Xu et al., 2025). While this line of research examines various interventions that are designed to directly influence prosocial behavior, there are growing attempts to identify factors that are seemingly irrelevant to prosocial behavior. For example, Hu et al. (2025) tests the unintended effects of a social antecedent (i.e., unearned preferential treatment) on prosocial behavior. Consistent with this literature, our research identifies a situational antecedent, namely robot prevalence, which can exert an unexpected positive impact on consumer prosocial behavior.

Additionally, this study enhances our understanding of how a threat imposed by the rise of service robots influences individuals' behaviors. Although service robots can pose both a realistic threat and a threat to human identity (i.e., symbolic threat) (Huang et al., 2021), past research mainly focuses on the effects of human identity/symbolic threat on consumer behavior (e.g., Xu et al., 2024). Only organizational behavior studies discuss the notion of realistic threat in the context of service robots (Tang et al., 2023a; 2023b; Yam et al., 2023). Echoing the previous literature, we find that robot prevalence triggers realistic threat, thereby driving consumers' prosocial behaviors, manifested by helping others, making donations, and engaging in pro-environmental behavior. We further test the robustness of the role of realistic threat in Study 3

and show that the more individuals are impacted by AI/robots (i.e., those who are employed and, especially, those who use AI/robots at work), the more prosocial they may become.

Accordingly, this research extends the theoretical boundaries of both intergroup threat theory (Stephan et al., 2015) and terror management theory (Greenberg et al., 1986). First, while prior research on intergroup threat has largely focused on conflicts within traditional human-centric contexts, such as those involving race and gender (e.g., Fan, 2024; Plant and Devine, 2003), our work investigates a contemporary intergroup dynamic: the conflict between humans as the ingroup and robots as a non-human outgroup (e.g., Jackson et al., 2020; Zhou et al., 2022). Likewise, we broaden the conceptualization of existential threats within terror management theory. This framework has been centered on threats from human actions like war (Guttmann, Gilboa, and Partouche-Sebban, 2021) or natural disasters (Bai and Chen, 2025). We extend this by identifying non-human agents—specifically, robots—as a novel source of such terror. Collectively, our findings demonstrate that competition with robots can constitute a realistic threat, prompting individuals to engage in prosocial behavior as a coping mechanism to reaffirm their human identity.

4.2. Practical implications

The hospitality service robot market is the second-largest sector in the service robot market, with strong double-digit growth rates expected on average each year from 2023 to 2026 (Müller et al., 2024). As automation increasingly pervades the service landscape, understanding the resulting psychological and behavioral shifts of consumers offers critical strategic value.

First, our findings indicate that as the prevalence of robot workforces in various fields becomes more apparent, consumers are likely to be more inclined towards prosocial behaviors. Importantly, evidence from our four experimental studies suggest this effect extends beyond monetary donations to include broader prosocial actions, such as pro-environmental behaviors and interpersonal helping. Therefore, hospitality firms operating in the era of robot prevalence should actively promote prosocial initiatives to boost customer participation and maximize social impact. Specifically, as consumers subconsciously seek to reaffirm their connection to the living world in response to robotic artificiality, they may be more responsive to prosocial initiatives. Hotels operating in high-automation settings via service robots could simultaneously emphasize eco-friendly practices, such as towel reuse program, carbon-offset options, or the sourcing local organic produce.

Furthermore, to maximize the efficiency of these initiatives, firms should adopt targeted segmentation strategies. The results of Study 3 suggest that prosocial campaigns are the most effective when targeting currently employed individuals, particularly those who are affected by AI/robots at work. For example, as low-skilled workers in sectors like manufacturing have been most affected by automation (Petrova et al., 2024), marketing communications directed at these segments might emphasize themes of community solidarity and mutual aid. Moreover, for higher-wage workers and those with higher education levels who tend to have higher levels of exposure to AI in daily tasks (Pew Research, 2023), hospitality firms could frame prosocial activities as opportunities for human connection to boost the efficiency of the campaigns.

Additionally, practitioners should consider geographical and demographic factors when designing prosocial activities to maximize consumer engagement. Our supplementary panel data analysis, using data from the International Federation of Robotics, provides empirical evidence for this, revealing a positive relationship between the prevalence of industrial robots and charitable contributions in U.S. metropolitan areas (see Appendix for details). The results indicate that metropolitan areas with a higher density of robots (e.g., South Bend, IN; Lafayette, IN; Toledo, OH) report greater charitable contributions than other areas with lower robot density (e.g., Anchorage, AK; Fairbanks, AK; Laredo, TX). This suggests that the broad societal trend of automation can generate tangible, real-world consequences for prosociality, offering a strategic insight for managers aiming to design more effective initiatives.

4.3. Limitations and directions for future research

First, we relied on online experiments to establish causality in the proposed effects. It is warranted to conduct a field experiment to examine individuals' actual prosocial behavior given a prevalent robot workforce.

Also, our experimental studies operationalized the prevalence of robots using physically embodied robots. However, non-embodied artificial intelligence (AI) is also pervasive in the hospitality industry, such as chatbots, mobile concierges, and AI systems used for dynamic pricing, marketing, and human resource operations. These non-physical forms of AI also shape the service environment and are potentially threatening humans' jobs. Hence, future research should examine whether the proposed effects can be replicated in the context of non-embodied AI.

Moreover, it is well-established that prosocial behavior is contingent upon various factors including personal values, economic factors, and cultural norms, which were not fully addressed in the present research. While our experimental designs applied in our study are able to help isolate the external factors' impact on prosocial behavior, future research could enhance robustness and generalizability by measuring and statistically controlling these potential factors.

Finally, the proposed effects of robot prevalence could be time-sensitive: people may perceive a greater realistic threat because AI and robots are in a relatively early stage, and major media coverage often focuses on how robots will replace humans (e.g., Jon, 2024). However, over time, as more people adopt AI/robots at work, they may be viewed more as useful tools than as potential competitors for jobs. Thus, our hypothesized effects may not occur in the future. It would be interesting to take a longitudinal approach and examine the effects of robot prevalence from a long-term perspective.

CRediT authorship contribution statement

Jihwan Yeon: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Sungwoo Choi:** Writing – original draft, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijhm.2026.104602.

Data availability

Data will be made available on request.

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